

Tightening Cycles and Inequality^{*}

Panagiotis Veneris[†]

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Abstract

This paper analyzes the interaction between credit tightening and inequality. In a standard New-Keynesian model with financial frictions and household heterogeneity I first show that asset purchases are expansionary and redistributive. To better understand monetary policy transmission, I decompose the aggregate responses into direct and indirect effects. Indirect effects are shown to play a dominant role in driving the economy's response during the build-up of the asset purchase program. I then explore an aggressive unwinding of the central bank's balance sheet and show this policy to be deflationary and dis-equalizing, imposing a trade-off for policymakers. While monetary policy can commit to deliver on its price stability mandate in response to supply side disruptions, this comes at the cost of weakening the equalizing forces of credit easing.

JEL Codes: D63, E32, E44, E52

Keywords: DSGE; Monetary Policy; Inequality

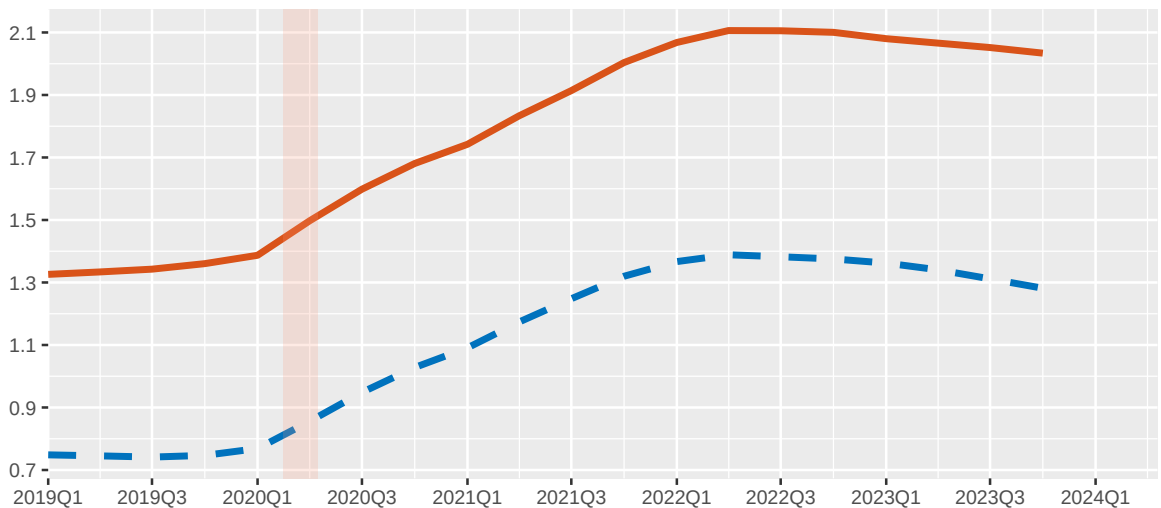
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[†]University of Liverpool Management School, Chatham Street, Liverpool, L69 7ZH, UK. Email: Panagiotis.Veneris@liverpool.ac.uk

1 INTRODUCTION

The Covid-19 pandemic posed a major challenge for advanced and emerging market economies, compromising financial stability mostly through disruptions in global supply chains with adverse spillovers in aggregate demand, causing a protracted supply-demand doom loop (Fornaro and Wolf, 2020). This brought monetary authorities, once again, in quest of a right policy mix to restore the missing inflation and stimulate economic activity. In March 2020, complementing a cut in the key policy rate, the European Central Bank (ECB) performed a new round of outright security purchases under the (already) established asset purchase program (APP), and the newly launched pandemic emergency purchase program (PEPP). Those purchases spanned a mix of private and government assets, from covered bonds (CBPP) and asset-backed securities (ABSPP) to public sector (PSPP) and corporate securities (CSPP). Figure 1 shows the evolution of the ECB's balance sheet (solid orange line), expanding from €1.32 trillion in Q1:2019 to over €2 trillion in Q1:2022. The ultimate scope of the expansion was to bolster aggregate demand, substituting away from services into traded goods. This is evident in the incremental path of securities held for monetary policy purposes (dashed blue line), which closely tracks the actual path of total assets, underscoring a deliberate objective to heat the economy.

Figure 1. ECB Balance Sheet (*Trillion €*)



NOTES: Total assets (solid orange line) and securities held for monetary policy purposes (dotted light-blue line) in the ECB balance sheet. Light pink shaded line shows the start of the pandemic emergency purchase program (PEPP), complementing the asset purchase program (APP). Source: ECB and author's calculations.

Relative to large-scale asset purchases, the pandemic emergency assistance took place with a substantially larger expansion in public debt issued by European sovereigns. Figure 2 quantifies the contribution of different assets in new net purchases under the PEPP and the APP following the Covid-19 outbreak, and highlights two key aspects of the programs. First, public sector purchases made up the majority of new securities on the augmented central bank balance sheet, playing a critical role in sustaining aggregate demand during the pandemic.¹ Second, effective 2023 the ECB unloaded its balance sheet by actively selling assets in the secondary market, a form of what is termed as *aggressive unwinding*.²

Figure 2. Net purchases by asset-category (*Billion €*)



NOTES: CB: corporate bonds (PEPP). CBND: covered bonds (PEPP). CBPP3: covered bonds (APP). CSPP: corporate sector securities (APP). PSPP: public sector securities (APP). PSS: public sector securities (PEPP). Source: ECB and author's calculations.

What are the implications of an aggressive tightening cycle for household inequality? Are there asymmetric effects across different agent cohorts and measures of inequality? If so, what are the underlying channels through which monetary policy propagates?

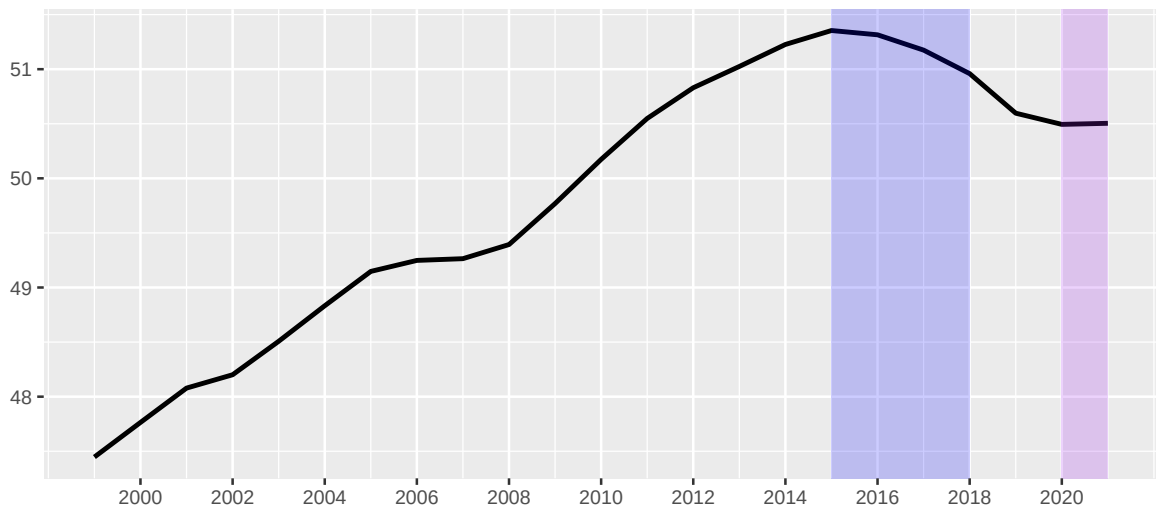
To answer these questions, I begin by establishing an empirical relation between central bank balance sheet policies and inequality based on historical events. Figure 3 provides historical evidence on the income inequality index for the euro area (E.A.). Income

¹This can be seen by contrasting the middle and right panels in the second row against the remaining panels in the figure.

²Airaudo (2023).

inequality is proxied by the Gini coefficient on the scale $[0, 100]$, with higher values reflecting a more unequal distribution. To construct the Gini index, I exploit data from the Standardized World Income Inequality Database (henceforth SWIID) focusing on equivalized market income.³ For a unified E.A. indicator I aggregate inequality data for all country members and adjust it by the respective GDP. As the figure suggests, income inequality in the E.A. followed a steady upward trend since its establishment. This trend persisted until the first quarter of 2015, at which point the European Central Bank initiated the first wave of asset purchases. The launch of the APP marked a prompt and secular decline in the area wide inequality index, rendering unconventional policies not only expansionary but also redistributive.

Figure 3. E.A. Gini index



NOTES: Source: Standardized World Income Inequality Database and author's calculations.

Practically, liquidity interventions have remained in effect almost since their debut, either in the form of new net purchases or in the form of reinvestments of principal payments from maturing securities. With the swift rebound in aggregate demand following the exit from the pandemic era, inflationary pressures emanating from ample monetary and fiscal injections pose a unique challenge for central banks. This necessitates policymakers to carefully re-design balance sheet interventions that align with the price stability mandate. The perils of this policy adjustment for economic activity and household inequality

³Solt (2020). Samarina and Nguyen (2024) discusses the merits of the SWIID database for a dynamic analysis of inequality.

remain elusive and warrant further investigation.⁴

Against this background, I propose a macroeconomic model of central bank intermediation and household heterogeneity to study the loading of the ECB’s balance sheet during the recent past and the joint dynamics of macroeconomic aggregates and inequality. I show that asset purchases when markets are destabilized can promote income and consumption convergence by affecting sources of household earnings asymmetrically across the income distribution. Specifically, open market operations tend to boost income and consumption for households in the lower tail of the distribution while dampening the effect for households in the right tail, thereby reducing overall skewness. In that sense, my model aligns with the empirical facts, confirming that expansionary unconventional policies contribute to reducing income and consumption inequality.

Building on these insights, I then proceed with a simulation that examines the extent to which a policy reversal — credit tightening — in light of the post-pandemic inflation surge reshapes the distributional income and consumption dynamics. I show that unwinding the central bank balance sheet is *deflationary* and *dis-equalizing*. Households that only consume according to disposable income are asymmetrically hit due to the economy’s slowdown, thereby partly undoing the positive effects of expansionary monetary policy on inequality.

In the model a subset of households is excluded from participating in asset markets and only consumes disposable income. The rest trade in markets for state-contingent securities and enjoy capital gains from the ownership of financial and non-financial entities, in addition to wage income. This distinction generates heterogeneous responses to policy shocks, contingent on the position of the household in the income distribution. The cyclicity of direct (partial equilibrium) and indirect (general equilibrium) effects plays a critical role in determining these differential responses, ultimately influencing the dynamics of household inequality.

Literature My paper relates to at least four broad strands of the literature. First, it builds on the extensive body of work on models with financial frictions, pioneered by [Bernanke et al. \(1999\)](#) and further developed by [Gertler and Kiyotaki \(2010\)](#) and [Gertler and Karadi \(2011\)](#). This literature highlights frictions in bank fund interme-

⁴A notable work in this direction is [Airaudo \(2023\)](#), who studies quantitative tightening policies under alternative monetary and fiscal regimes.

diation, which endogenously constrain financial intermediaries' access to credit, thereby amplifying macroeconomic shocks. Several important contributions expand on this framework. [Gertler and Karadi \(2013\)](#) and [Sims and Wu \(2021\)](#) examine the macroeconomic effects of central bank interventions in private and government asset markets, while [Gertler and Kiyotaki \(2015\)](#) explore the interaction between financial frictions and bank runs. [Cúrdia and Woodford \(2016\)](#) analyze optimal monetary policy in the presence of credit constraints. The present paper can be considered as part of this line of research, with a special interest on how financial frictions shape unconventional policies, and the implications of the latter for household inequality.

Second, my paper relates to the pool of literature on tractable heterogeneous agents models, mostly represented by the Two-Agent New Keynesian (TANK) framework. As emphasized in [Kaplan et al. \(2018\)](#), household heterogeneity is an important determinant of the power of the monetary transmission mechanism, and therefore should be taken into account when analyzing monetary policy. Tractable heterogeneous agent models (mostly TANK) have been used extensively since the influential works of [Galí et al. \(2007\)](#) and [Bilbiie \(2008\)](#). [Bilbiie et al. \(2013\)](#) evaluate the redistributive and aggregate effects of alternative fiscal policies in a setup with constrained and unconstrained households —a theme also explored by [Cantore and Freund \(2021\)](#) within a worker-capitalist framework. More recently, [Debortoli and Galí \(2024\)](#) assess the extent to which the TANK model can approximate reasonably well the aggregate dynamics delivered by the heterogeneous agent Bewley-type (HANK) counterparts. The present paper contributes to this line of work by incorporating a canonical retail banking sector subject to agency frictions within a TANK framework to study the interaction between unconventional policy and inequality.

Third, my paper engages with the broad literature assessing the macroeconomic stabilization effects of large-scale asset purchases. Several influential studies have analyzed such policies within closed-economy settings, including [Gertler and Karadi \(2011\)](#), [Chen et al. \(2012\)](#), [Gertler and Karadi \(2013\)](#), [Carlstrom et al. \(2017\)](#), and [Sims and Wu \(2021\)](#). In contrast, [Dedola et al. \(2013\)](#) and [Alpanda and Kabaca \(2020\)](#) examine the international transmission of unconventional policies using open-economy frameworks. As is widely discussed in those theoretical works, large scale asset purchases programs (LSAPS) have real effects by suppressing term and risk premia through liquidity in-

jections. Furthermore, empirical evidence supports the countercyclical implications of LSAPS on long term yields of different assets. Employing an event-study methodology, [Vissing-Jorgensen and Krishnamurthy \(2011\)](#) conclude that the quantitative easing programs implemented by the Federal Reserve effectively lowered the yields of assets mainly purchased in bulk during each of the three phases (QE1, QE2, QE3). [D’Amico and King \(2013\)](#) reach similar conclusions by estimating the effects of a treasury supply shock on yield curve dynamics, and [Gagnon et al. \(2011\)](#) focus on portfolio balance effects in flattening the yield curve. I contribute to this literature by providing a thorough analysis of the mechanics of central bank intermediation, zooming in on the responses of public and private premia and the spillovers to all related variables, rather than providing a high-level overview common in the previous analyses —some of which entirely abstract from key macroeconomic aggregates.⁵

Fourth, and most importantly, my paper expands the literature on monetary policy and inequality. The literature so far offers mixed evidence with regards the effects of monetary policy on inequality. On the one hand, a number of papers find that expansionary conventional and unconventional monetary policy reduces income and consumption inequality ([Furceri et al., 2018](#); [Hohberger et al., 2020](#); [Komatsu, 2023](#); [Samarina and Nguyen, 2024](#); [Lenza and Slacalek, 2024](#)). Others favor the clear opposite ([Saiki and Frost, 2014](#); [Mumtaz and Theophilopoulou, 2017](#); [Montecino et al., 2017](#); [Dolado et al., 2021](#)). In this paper, I contribute to this debate by showing that accommodative monetary policy reduces income and consumption inequality. Moreover, and while most existing research examines how conventional and unconventional monetary easing influences inequality, this paper investigates the distributional effects of a monetary tightening. Specifically, I analyze how a credit tightening impacts income, consumption, and wealth inequality, thereby offering new insights into the asymmetric effects of monetary policy.

⁵For example, somewhat paradoxically [Chen et al. \(2012\)](#) and [Gertler and Karadi \(2013\)](#) abstract from the consumption response to open market operations by the central bank.

2 MODEL

The model builds on the textbook New Keynesian framework augmented with financial frictions (Gertler and Kiyotaki, 2010; Gertler and Karadi, 2011). To this, I introduce household heterogeneity by assuming two household types: unconstrained (optimizing) households are free to borrow and lend to smooth consumption over time; credit constrained (hand-to-mouth) households are entirely excluded from financial markets. This distinction generates variations in the consumption, saving, and labor supply decisions across cohort types in response to exogenous aggregate shocks, allowing to study the nexus between (unconventional) central bank balance sheet policies and inequality.

There are six agents in the model: (i) households, (ii) labor unions, (iii) firms, (iv) financial intermediaries (henceforth banks), (v) a monetary authority (central bank), (vi) and a fiscal authority (Treasury). Households have access to three types of assets: short-term deposits D_t ; real balances of long-term government bonds $q_t B_t$ (B_t are units of long term bonds each priced at the real price q_t) that pay a constant (real) coupon ν ; and privately intermediated capital claims S_t issued at the (real) price Q_t .

Banks act as intermediaries between households, productive firms, and the government. As is standard in the literature that assesses the effects of unconventional policies, banks are subject to a moral hazard friction that introduces an (endogenous) incentive-compatibility constraint. This constraint limits the optimal amount of bank leverage, breaks Wallace (1981) neutrality, and gives role to quantitative easing (tightening) policies.⁶

In the model, moral hazard interacts with a variety of real and financial frictions. On the demand side, households face a quadratic portfolio adjustment cost when adjusting asset portfolios away from a frictionless benchmark. On the supply side, labor unions and firms are subject to Rotemberg (1982) adjustment costs when varying labor wages and product prices away from the (price) inflation target. Those frictions help the model generate the main macroeconomic effects of balance sheet policies while matching key empirical moments.⁷

⁶Other frictions that can break the Wallace (1981) neutrality is the assumption of segmented asset markets (Carlstrom et al., 2017), or the introduction of portfolio adjustment costs (Hohberger et al., 2019).

⁷For example, wage adjustment costs support the existence of procyclical profits in New Keynesian models conditional on monetary policy shocks.

I close the model with two central authorities. A central bank conducts conventional interest rate policy in normal times and expands its toolkit with unconventional measures during periods of financial stress. The Treasury issues long term bonds to finance public spending, and levies lump-sum taxes to balance its budgetary rule.

2.1 Long term bonds

Long term bonds follow the modeling convention in [Hatchondo and Martinez \(2009\)](#). A government bond B_t issued in period t at price q_t is a security that obligates the issuer to a sequence of geometrically decaying coupon payments $\nu, (1 - \varrho)\nu, (1 - \varrho)^2\nu \dots$ where $\varrho \in [0, 1]$ controls the decay parameter for cash flows and reflects the maturity structure of government debt; in the two polar cases, $\varrho = 1$ corresponds to short term debt (one period maturity), and $\varrho = 0$ represents a perpetuity (consol). The attractive feature of long term bonds is that current bond issuances are sufficient to infer the future stream of debt obligations stemming from past issuances. Formally, I can express this as

$$B_{t-j} = \sum_{j=1}^{\infty} (1 - \varrho)^j B_t.$$

A bond issued in period $t - j$ is equivalent to $(1 - \varrho)^j$ bonds issued in period t at price q_t , that pay a sequence of known future cash flows. In that sense, B_{t-j} represents long term debt in equivalent newly issued long term bonds B_t . All future payment obligations are summarized in a single state variable, B_{t-j} .

2.2 Households

There is a continuum of infinitely lived households in the unit interval. All households are assumed to have identical preferences, given by the instantaneous utility function

$$\ln(c_{t+i}^j) - \kappa_h \frac{(h_{t+i}^j)^{1+\varphi}}{1+\varphi}. \quad (1)$$

c_{t+i}^j stands for the consumption per capita, and h_{t+i}^j is the labor supply. The super-index $j \in \{o, r\}$ specifies the household type; for $j = o \in [0, 1 - \eta]$ I refer to the optimizing households, and for $j = r \in [0, \eta]$ to the liquidity constrained (hand-to-mouth) households. There is perfect consumption risk sharing within each group. Households

discount expected future utility with the discount factor $\beta \in (0, 1)$. The inverse of the elasticity of labor supply to variations in the wage rate is given by φ , and $\kappa_h > 0$ is a scaling parameter that captures the weight on the disutility from labor.

Optimizing households A share $(1 - \eta)$ of households are optimizers in that they behave in a Ricardian fashion and consume taking into account the intertemporal budget constraint. Optimizing households are forward looking and smooth consumption by trading in complete markets for non-contingent securities. Their asset portfolio consists of short term bank deposits d_t^o , long term government bonds B_t^o , and corporate (firm) shares S_t^o . I follow [Gertler and Karadi \(2013\)](#) by assuming that households are inefficient investors that incur a transaction cost when holding bonds and shares above or below a certain threshold.⁸ The threshold level for bonds and shares, $\bar{B}^o$ and $\bar{S}^o$, respectively, is the frictionless capacity of each asset that unconstrained households can hold in their portfolio without requiring any broker expertise. Deviations of B_t^o and S_t^o from their respective target levels $\bar{B}^o$ and $\bar{S}^o$ are subject to a quadratic portfolio adjustment cost with a common adjustment parameter κ . Portfolio adjustment costs are widely used in models with unconventional central bank interventions as they break the arbitrage condition between assets and give role to meaningful policy implications ([Gertler and Karadi, 2013](#); [Harrison, 2017](#); [Hohberger et al., 2019](#); [Guo et al., 2023](#)).⁹

The budget constraint of optimizing households can be written as

$$\begin{aligned} c_t^o + d_t^o + q_t \left[B_t^o + \frac{1}{2} \kappa (B_t^o - \bar{B}^o)^2 \right] + Q_t \left[S_t^o + \frac{1}{2} \kappa (S_t^o - \bar{S}^o)^2 \right] + \frac{\gamma_w}{2} (\pi_t^w - \pi)^2 Y_t + t_t^o \\ \leq W_t h_t^o + Div_t^o + R_t^d d_{t-1}^o + R_t^b q_{t-1} B_{t-1}^o + R_t^k Q_{t-1} S_{t-1}^o + \Gamma_t^o. \end{aligned} \quad (2)$$

Optimizing households consume (c^o), deposit to the banks (d^o), invest in long term government bonds (B^o) and private capital claims (S^o), pay lump sum taxes (t^o) and wage adjustment costs.¹⁰ They receive labor income (Wh^o), interest income on deposits ($R^d d^o$), government bonds ($R^b q B^o$), and private capital claims ($R^k Q S^o$). W is the real wage, and

⁸One can think of portfolio costs as capturing a form of asset market segmentation, whereby households are not specialists in this segment of the market and face transaction costs when investing in long term bonds ([Chen et al., 2012](#)). Transaction costs prevent optimizers from fully exploiting arbitrage opportunities between short term and long term bonds.

⁹Portfolio costs help the model deliver positive term spreads on long term bonds.

¹⁰Details on the adjustment cost function are found in [subsection 2.3](#).

R^d , R^b , R^k the respective real gross returns on deposits, long term government bonds, and firms' capital claims. Div^o are profits accrued to optimizers (from ownership of intermediate and capital producing firms) net of transfers to rule of thumb households, and Γ^o are stochastic dividend payouts from the ownership of financial intermediaries. It is worth noting that d^o is the sum of private deposits and deposits created by the exchange of securities (be it private or government) with reserves when the central bank intervenes in asset markets.

My setup shares similarities with [Gertler and Karadi \(2011\)](#) and [Gertler and Karadi \(2013\)](#). For instance, I assume that deposits can be thought of as one period riskless assets that pay the gross risk free rate, i.e. $R_t^d = R_t$, where R_t is the gross real risk free rate. Hence, deposits and short term government bonds are perfect substitutes. A main difference of my model is that I allow unconstrained households to directly hold both short and long term debt.

The problem of optimizing households is to choose the sequence $\{c_{t+i}^o, d_{t+i}^o, B_{t+i}^o, S_{t+i}^o\}_{i=0}^{\infty}$ to maximize lifetime utility $E_t \sum_{i=0}^{\infty} \beta^i \left[\ln(c_{t+i}^o) - \kappa_h \frac{(h_{t+i}^o)^{1+\varphi}}{1+\varphi} \right]$ subject to the budget constraint. E_t is the expectations operator conditional on time t information.

The first order conditions are

$$1 = E_t \Lambda_{t,t+1}^o R_{t+1} \quad (3)$$

$$\Lambda_{t,t+1}^o = \beta \frac{c_t^o}{c_{t+1}^o} \quad (4)$$

$$B_t^o = \bar{B}^o + E_t \Lambda_{t,t+1}^o \frac{(R_{t+1}^b - R_{t+1})}{\kappa} \quad (5)$$

$$S_t^o = \bar{S}^o + E_t \Lambda_{t,t+1}^o \frac{(R_{t+1}^k - R_{t+1})}{\kappa}. \quad (6)$$

Equation (3) is the first order condition for deposits. Equation (4) denotes optimizers' stochastic discount factor in period t of expected future $(t+1)$ real payoffs. Equations (5) and (6) are the respective optimal choices for long term government bonds and private capital claims priced using the optimizers' intertemporal marginal rate of substitution.

The optimal decisions for bonds and private assets highlight a relation between the demand for those securities, the excess returns and the transaction costs. Demand for securities above the frictionless level is increasing with the excess returns relative to the portfolio cost parameter. When the portfolio cost parameter goes to infinity, households

demand exactly the frictionless capacity of each asset. In contrast, when the transaction cost is zero, households can engage in frictionless arbitrage between the two assets, and the term premium disappears.¹¹

Constrained households Liquidity constrained households on the $[0, \eta]$ interval are entirely excluded from asset markets (i.e. credit constrained) and just consume disposable income every period. They hold neither securities (private or government) nor deposits to smooth consumption in response to variations in labor income. In that sense they face a static problem and behave in a "hand-to-mouth" fashion (Galí et al., 2007). I further assume that constrained households are not taxed, but instead receive lump sum transfers tr_t^r from the government.

The problem of an individual constrained household is to choose consumption to maximize period utility

$$\ln(c_t^r) - \kappa_h \frac{(h_t^r)^{1+\varphi}}{1+\varphi}, \quad (7)$$

subject to the (Keynesian) budget constraint

$$c_t^r + \frac{\gamma_w}{2} (\pi_t^w - \pi)^2 Y_t \leq W_t h_t^r + tr_t^r. \quad (8)$$

Utility maximization implies that (8) holds with equality.

2.3 Wage setting

Wage setting decisions are made by a continuum of labor unions $\zeta \in [0, 1]$, each of which represents workers of type ζ . Firms regard each type of labor service ζ as an imperfect substitute for the labor services of other households. Unions operate in a monopolistically competitive labor market and set the (real) wage $w_{\zeta,t}$ to maximize worker's ζ utility. This creates (real) wage rigidities. Apart from the empirical support, wage rigidities favor the existence of procyclical profits conditional on monetary policy shocks, a feature orthogonal to the implications of the standard flexible-wage New Keynesian model.¹²

¹¹The term premium (or bond spread) refers to the difference between the expected return on the long term government bond and the risk free rate.

¹²An important implication of the labor market structure is that labor supply decisions are no longer endogenous choice variables for households. Instead, assuming that their representation in the labor market is exclusively delegated to labor unions who decide —on their behalf— the real wage, given the prevailing wage firms choose how much labor to employ and households stand ready to supply as many hours demanded by firms.

Households are assumed to be uniformly distributed across labor unions (Galí et al., 2007; Colciago, 2011; Ascari et al., 2017). This implies that each union ζ represents an equal share of η liquidity constrained and $(1-\eta)$ optimizing households. I further assume that firms are indifferent with respect to labor type ζ they employ. Hence, aggregate demand by firms for labor type ζ is allocated uniformly between unions. It follows that all households supply the same amount of hours worked, and in the aggregate it holds that $h_{\zeta,t}^o = h_{\zeta,t}^r \equiv H_{\zeta,t}$. This implies that all unions ζ supply the same hours $H_{\zeta,t}$ to firms.

Each period, labor union ζ chooses the (real) wage $w_{\zeta,t}$ to maximize the objective function

$$\eta \left[\frac{1}{c_{\zeta,t}^r} w_{\zeta,t} H_{\zeta,t} - \kappa_h \frac{H_{\zeta,t}^{1+\varphi}}{1+\varphi} \right] + (1-\eta) \left[\frac{1}{c_{\zeta,t}^o} w_{\zeta,t} H_{\zeta,t} - \kappa_h \frac{H_{\zeta,t}^{1+\varphi}}{1+\varphi} \right] \quad (9)$$

subject to the demand for individual labor

$$H_{\zeta,t} = \left(\frac{w_{\zeta,t}}{W_t} \right)^{-\epsilon_w} H_t. \quad (10)$$

Because consumption will differ between the two types of households, the union weighs labor income with the respective marginal utilities, where $1/c_{\zeta,t}^r$ and $1/c_{\zeta,t}^o$ is the marginal utility of consumption of liquidity constrained and unconstrained workers of type ζ , respectively.

All unions face the same demand for their workers, and set the same (aggregate) real wage. Imposing symmetry in the labor market, the first order condition becomes

$$\left(\frac{\eta}{\kappa_h c_t^r H_t^\varphi} + \frac{1-\eta}{\kappa_h c_t^o H_t^\varphi} \right) W_t = \mu_w. \quad (11)$$

W_t is the aggregate (real) wage index, H_t the aggregate labor supply, and $\mu_w = \epsilon_w / \epsilon_{w-1}$ the real wage markup, where ϵ_w stands for the elasticity of substitution between differentiated labor services.

Following Hall (2005) and Blanchard and Galí (2007) I introduce rigidities in the labor market. Labor unions face wage setting frictions in that they have to pay a quadratic cost (Rotemberg, 1982) in the deviation of real wage from the (price) inflation target (Fernández-Villaverde et al., 2015).

The real wage adjustment cost function takes the following form

$$AC_{\zeta,t}^w = \frac{\gamma_w}{2} \left(\frac{w_{\zeta,t}}{w_{\zeta,t-1}} - \pi \right)^2 Y_t,$$

where γ_w is a parameter governing the degree of the adjustment cost, π is steady state price inflation, and Y_t is the aggregate output. In a symmetric equilibrium, the problem of a typical labor union ζ is to choose the real wage $w_{\zeta,t}$ to maximize a weighted average of the utility of its members, subject to the associated budget constraints. Optimization yields the New Keynesian Wage Phillips Curve (*NKWPC*)

$$(1 - \eta) \frac{1}{c_t^o} \left[(\epsilon_w - 1) W_t H_t + \gamma_w (\pi_t^w - \pi) \pi_t^w Y_t - \gamma_w E_t \Lambda_{t,t+1}^o (\pi_{t+1}^w - \pi) \pi_{t+1}^w Y_{t+1} \right] + \eta \frac{1}{c_t^r} \left[(\epsilon_w - 1) W_t H_t + \gamma_w (\pi_t^w - \pi) \pi_t^w Y_t - \gamma_w E_t \Lambda_{t,t+1}^r (\pi_{t+1}^w - \pi) \pi_{t+1}^w Y_{t+1} \right] = \epsilon_w \kappa_h H_t^{1+\varphi}. \quad (12)$$

$\Lambda_{t,t+1}^r$ is the stochastic discount factor of constrained households, and π^w is the real wage inflation.

I follow [Erceg et al. \(2000\)](#) and [Sims and Wu \(2021\)](#) and postulate that there exists a perfectly competitive labor aggregator (labor packer). The labor aggregator buys differentiated labor services ($H_{\zeta,t}$) given the wages ($w_{\zeta,t}$) set by the unions, and aggregates those labor inputs into a single labor index H_t , which then leases to firms at the (real) aggregate wage index W_t . Aggregate labor can be expressed as a constant elasticity of substitution (CES) aggregator of heterogeneous labor types $H_t = \left(\int_0^1 H_{\zeta,t}^{\frac{\epsilon_w-1}{\epsilon_w}} dj \right)^{\frac{\epsilon_w}{\epsilon_w-1}}$. The problem of the labor packer is to choose the quantity of labor type ζ employed to solve $\max_{H_{\zeta,t}} W_t \left(\int_0^1 H_{\zeta,t}^{\frac{\epsilon_w-1}{\epsilon_w}} d\zeta \right)^{\frac{\epsilon_w}{\epsilon_w-1}} - \int_0^1 w_{\zeta,t} H_{\zeta,t} d\zeta$. Optimization yields the total demand for type ζ labor, $H_{\zeta,t} = \left(\frac{w_{\zeta,t}}{W_t} \right)^{-\epsilon_w} H_t$. Finally, exploiting the zero profit condition, the aggregate real wage index follows $W_t = \left(\int_0^1 w_{\zeta,t}^{1-\epsilon_w} dj \right)^{\frac{1}{1-\epsilon_w}}$.¹³

Aggregation Aggregate variables are defined as the sum of individual variables, ad-

¹³Note that the aggregate real wage index is taken as given by the labor packer.

justed by the share of each household type in the population

$$C_t \equiv (1 - \eta)c_t^o + \eta c_t^r \quad (13)$$

$$H_t \equiv (1 - \eta)h_t^o + \eta h_t^r \quad (14)$$

$$T_t \equiv (1 - \eta)t_t^o \quad (15)$$

$$\Gamma_t \equiv (1 - \eta)\Gamma_t^o \quad (16)$$

$$tr_t \equiv \eta tr_t^r. \quad (17)$$

Assets and dividend payments held within the household sector are indexed by H and expressed as

$$B_t^H \equiv (1 - \eta)B_t^o \quad (18)$$

$$S_t^H \equiv (1 - \eta)S_t^o \quad (19)$$

$$D_t^H \equiv (1 - \eta)d_t^o \quad (20)$$

$$Div_t^H \equiv (1 - \eta)Div_t^o \quad (21)$$

2.4 Financial intermediaries

A continuum of imperfectly competitive banks spans the unit interval $f \in [0, 1]$. Bank f issues non-contingent riskless short term debt (deposits) to households, $d_{f,t}^H$, supplies credit to non financial firms in exchange for capital claims, $S_{f,t}^B$ (priced at Q_t), invests in long term government bonds, $B_{f,t}^B$ (priced at q_t), and exchanges securities for central bank reserves, $m_{f,t}^B$, when the central bank expands its balance sheet.

Let the amount of own equity (net worth) be $n_{f,t}$ at the end of period t . The bank's balance sheet becomes

$$Q_t S_{f,t}^B + q_t B_{f,t}^B + m_{f,t}^B = n_{f,t} + d_{f,t}^H. \quad (22)$$

Over time, accumulated net worth evolves as the difference between interest payments on assets and liabilities (i.e. accumulated retained earnings)

$$n_{f,t+1} = R_{t+1}^k Q_t S_{f,t}^B + R_{t+1}^b q_t B_{f,t}^B + R_{t+1}^m m_{f,t}^B - R_{t+1} d_{f,t}^H, \quad (23)$$

where R_t^m stands for the real gross return on central bank reserves (i.e. reserve rate).

If I let Z_t be the net income accruing to banks from holding firm equity, and δ the capital depreciation rate, the stochastic gross real rate of return on the loan (loan rate) can be defined as

$$R_{t+1}^k = \frac{Z_{t+1} + (1 - \delta)Q_{t+1}}{Q_t}. \quad (24)$$

Beyond firm equity, bank f invests in long term government bonds. In its general form, the real gross return on the long term government bond (bond rate) issued at price q_t can be expressed as $R_{t+1}^b = (\nu + (1 - \varrho)q_{t+1})/q_t$. To simplify computations, I assume that the long term bond is a consol (i.e. $\varrho = 0$) that pays every period a constant coupon ν normalized to a unit of the final good.¹⁴ As a consequence, the bond rate reduces to

$$R_{t+1}^b = \frac{1 + q_{t+1}}{q_t}. \quad (25)$$

Banks are owned by households (optimizers). The expected time horizon of each bank is finite, and a bank can exit the banking industry each period with probability $(1 - \sigma_B)$, $\sigma_B \in [0, 1)$. This is to ensure that banks do not retain enough net worth to self-finance investment projects, thereby relaxing any financial constraints.

Bank's problem. The problem of each individual bank f is to maximize the present value of expected future dividends

$$V_{f,t} = E_t \sum_{i=0}^{\infty} (1 - \sigma_B) \sigma_B^i \left[\Lambda_{t,t+i+1}^o n_{f,t+i+1} \right], \quad (26)$$

where banks discount future streams of net worth using the optimizer's stochastic discount factor. As in [Gertler and Kiyotaki \(2010\)](#) and [Gertler and Karadi \(2011\)](#) I introduce a principal agent friction; every period, the bank is tempted to run away with a fraction of assets for personal gain. In this case, the bank exits the industry. This imposes a limit on the amount of deposits that households are willing to hold within the bank. Specifically, households roll over deposits only to the extent that the franchise value of the bank is greater than the gains from diverting assets

$$V_{f,t} \geq \theta \left[Q_t S_{f,t}^B + \Delta q_t B_{f,t}^B + \omega m_{f,t}^B \right]. \quad (27)$$

¹⁴A similar assumption is made in [Gertler and Karadi \(2013\)](#).

Equation (27) can be thought of as an incentive-compatibility constraint for it makes the bank indifferent between operating honestly and diverting assets, in the margin. In the equilibrium I study, I impose that (27) holds with equality. Assuming that $\Delta, \omega \in [0, 1]$, $\Delta\theta$ and $\omega\theta$ are the corresponding fractions of government bonds and central bank reserves the bank can divert, while $\theta \in [0, 1]$ is the fraction of private capital claims that can be absconded. The idea is that it is easier for a household to monitor the contribution of government bonds and central bank reserves to the intermediary's portfolio. Subsequently, it makes it harder for a bank to divert those assets relative to private capital claims. It follows that $\Delta\theta < \theta$ and $\omega\theta < \theta$.

The franchise value of the bank can be expressed recursively as a Bellman equation

$$V_{f,t}(S_{f,t}^B, B_{f,t}^B, m_{f,t}^B, d_{f,t}^H) = \max E_t \left\{ (1 - \sigma_B) \Lambda_{t,t+1}^\circ n_{f,t+1} + \sigma_B \Lambda_{t,t+1}^\circ V_{f,t+1}(S_{f,t+1}^B, B_{f,t+1}^B, m_{f,t+1}^B, d_{f,t+1}^H) \right\}. \quad (28)$$

Each period the bank chooses the sequence $\{S_{f,t}^B, B_{f,t}^B, m_{f,t}^B, d_{f,t}^H\}_{t=0}^\infty$ to maximize (28) subject to the balance sheet (22) and the incentive-compatibility constraint (27).

Guess that the value function has a linear form (Gertler and Kiyotaki, 2010)

$$V_{f,t}(S_{f,t}^B, B_{f,t}^B, m_{f,t}^B, d_{f,t}^H) = v_t^s S_{f,t}^B + v_t^b B_{f,t}^B + v_t^m m_{f,t}^B - v_t^d d_{f,t}^H, \quad (29)$$

where v_t^s is the marginal value of extending credit to non financial firms by one unit, v_t^b the marginal value of holding an extra unit of long term government bonds, v_t^m the marginal value of reserve holdings within the central bank, and v_t^d the marginal cost of deposits.

The optimality conditions are given by

$$E_t \Lambda_{t,t+1}^\circ \Omega_{t+1} (R_{t+1}^k - R_{t+1}) = \frac{\nu_t^{IC}}{1 + \nu_t^{IC}} \theta \quad (30)$$

$$E_t \Lambda_{t,t+1}^\circ \Omega_{t+1} (R_{t+1}^b - R_{t+1}) = \frac{\nu_t^{IC}}{1 + \nu_t^{IC}} \theta \Delta \quad (31)$$

$$E_t \Lambda_{t,t+1}^\circ \Omega_{t+1} (R_{t+1}^m - R_{t+1}) = \frac{\nu_t^{IC}}{1 + \nu_t^{IC}} \theta \omega, \quad (32)$$

where $\nu_t^{IC} \geq 0$ is the Lagrange multiplier on the incentive constraint. Combining (30)

and (31) yields an arbitrage condition between private and government assets

$$E_t \Lambda_{t,t+1}^o \Omega_{t+1} (R_{t+1}^b - R_{t+1}) = \Delta E_t \Lambda_{t,t+1}^o \Omega_{t+1} (R_{t+1}^k - R_{t+1}). \quad (33)$$

Focusing on an equilibrium in which the incentive constraint always binds, I can express the leverage constraint as the ratio of bank's intermediated assets over net worth¹⁵

$$\phi_{f,t} = \frac{Q_t S_{f,t}^B + \Delta q_t B_{f,t}^B + \omega m_{f,t}^B}{n_{f,t}}. \quad (34)$$

ϕ_f is the maximum amount of leverage consistent with the incentive constraint, and is defined as¹⁶

$$\phi_t \equiv \frac{E_t \Lambda_{t,t+1}^o \Omega_{t+1} R_{t+1}}{\theta - E_t \Lambda_{t,t+1}^o \Omega_{t+1} (R_{t+1}^k - R_{t+1})}, \quad (35)$$

where Ω defines the marginal value of bank's net worth according to

$$\Omega_t \equiv (1 - \sigma_B) + \sigma_B E_t \left\{ \Lambda_{t,t+1}^o \Omega_{t+1} \left[(R_{t+1}^k - R_{t+1}) \phi_t + R_{t+1} \right] \right\}. \quad (36)$$

Equations (30) and (31) characterize the excess returns on private and government assets, respectively, while (32) the excess returns on central bank reserves. Note that if the incentive constraint binds ($\nu_t^{IC} > 0$), limits to arbitrage drive excess returns of private capital claims, long term government bonds, and central bank reserves over the risk free rate. When the constraint does not bind ($\nu_t^{IC} = 0$), asset markets become frictionless and banks accumulate assets up the point where excess returns vanish. In the equilibrium I consider, I assume that banks can not divert central bank reserves for personal gain ($\omega = 0$).¹⁷ Hence reserves become riskless assets, and $R_t^m = R_t$.

Equation (34) highlights the role of the financial friction in generating an endogenous leverage constraint. When the incentive constraint binds, (34) ties leverage to net worth which evolves endogenously over time. It also stresses the relative importance of asset-specific central bank purchases in relaxing the leverage constraint; since $\omega < \Delta < 1$, the exchange of central bank reserves with securities when the central bank purchases assets relaxes the leverage constraint and promotes credit supply to non financial firms.

¹⁵The numerator is an adjusted measure of the maximum amount of assets a bank can hold that makes it respect (i.e. not becoming better-off by violating) the incentive constraint.

¹⁶Given the linearity of the value function, I can drop the f -index as leverage does not depend on bank-specific characteristics.

¹⁷In line with Sims and Wu (2021), and Coenen et al. (2018).

Finally, equation (35) balances the opportunity cost of cheating with that of operating honestly. It states that the willingness of households to roll over deposits is increasing in excess returns on private assets, and decreasing in the amount of assets that can (potentially) be absconded.

Aggregation Since ϕ_t does not depend on bank specific characteristics, I can aggregate across individual banks to get the aggregate leverage ratio in terms of (aggregate) net worth

$$\phi_t = \frac{Q_t S_t^B + \Delta q_t B_t^B + \omega M_t^B}{N_t}. \quad (37)$$

It follows that the aggregate balance sheet is

$$Q_t S_t^B + q_t B_t^B + M_t^B = N_t + D_t^H. \quad (38)$$

As I mentioned earlier, banks exit the industry every period with a stochastic *i.i.d* probability $(1 - \sigma_B)$. To keep the mass of operating banks constant over time, I assume that for every exiting bank there exists a new bank that enters the system. Each new bank receives startup funds.¹⁸ Those funds are fixed and amount to a fraction $\xi/(1 - \sigma_B)$ of the final period assets of exiting bankers. The aggregate net worth is divided between the net worth of surviving banks (denoted by the subscript s)

$$N_{s,t} = \sigma_B \left[R_t^k Q_{t-1} S_{t-1}^B + R_t^b q_{t-1} B_{t-1}^B + R_t^m M_{t-1}^B - R_t D_{t-1}^H \right] \quad (39)$$

and that of new banks (denoted by subscript y)

$$N_{y,t} = \frac{\xi}{(1 - \sigma_B)} (1 - \sigma_B) \left[Q_{t-1} S_{t-1}^B + q_{t-1} B_{t-1}^B + M_{t-1}^B \right]. \quad (40)$$

Combining (38), (39), and (40) implies an expression for the law of motion of aggregate net worth which is increasing in the excess returns to private ($R_t^k - R_t$) and government

¹⁸This is to ensure that banks have some "skin in the game", i.e. they can fund part of their business operations with own net worth.

$(R_t^b - R_t)$ assets

$$N_t = \sigma_B \left[(R_t^k - R_t) \frac{Q_{t-1} S_{t-1}^B}{N_{t-1}} + (R_t^b - R_t) \frac{q_{t-1} B_{t-1}^B}{N_{t-1}} + R_t \right] N_{t-1} + \xi \left[Q_{t-1} S_{t-1}^B + q_{t-1} B_{t-1}^B + M_{t-1}^B \right]. \quad (41)$$

2.5 Central authorities

Central Bank The central bank adjusts the short term (nominal) interest rate through an endogenous feedback rule similar to [Taylor \(1993\)](#) with interest rate smoothing ([Clarida et al., 2000](#)), to target deviations of inflation and output from the steady state

$$\frac{R_t^{nom}}{R^{nom}} = \left(\frac{R_{t-1}^{nom}}{R^{nom}} \right)^{\rho_m} \left[\left(\frac{\pi_t}{\pi} \right)^{\phi_\pi} \left(\frac{Y_t}{Y} \right)^{\phi_y} \right]^{1-\rho_m} e^{(\sigma_m \epsilon_m)}. \quad (42)$$

R_t^{nom} is the short term nominal rate, $0 \leq \rho_m < 1$ governs the degree of interest rate smoothing, $\phi_\pi > 1$ and $\phi_y \geq 0$ are the respective inflation and output gap coefficients, $\epsilon_m \sim N(0, \sigma_m^2)$ is the monetary policy innovation, and σ_m the standard deviation of innovation. Variables without time subscripts hold in steady state. The feedback rule implies that during normal times the central bank uses the short term nominal rate as its primary policy instrument. Real and nominal rates are linked together through a Fisher equation

$$R_t = \frac{R_{t-1}^{nom}}{\pi_t}.$$

Hence, in equilibrium, the nominal rate equals the inflation-adjusted deposit rate, such that $R_{t-1}^{nom} = R_t^d \pi_t$.

Beyond conventional interest rate policy, the central bank purchases long term government bonds and private securities at the respective prices, Q_t and q_t . Those securities are held by financial intermediaries and households. To offset the mismatch on its balance sheet, the central bank settles asset purchases with an equally sizable issuance of central bank reserves M_t^B .¹⁹ The central bank balance sheet expands; reserves appear directly on the budget constraint of financial intermediaries, and indirectly on optimizers' budget constraint through deposits. This is because when the central bank purchases

¹⁹For simplicity, I abstract from costs associated with central bank asset purchases, as highlighted in [Kurtzman and Zeke \(2020\)](#).

securities from households it must create liabilities (deposits). It does so by mobilizing financial intermediaries who credit the exact same amount on optimizers' deposit account in exchange for central bank reserves.

I can express the aggregate supply of central bank reserves as

$$M_t^B = Q_t S_t^{CB} + q_t B_t^{CB}. \quad (43)$$

The budget constraint of the central bank is given by

$$T_t^{CB} + R_t M_{t-1}^B + Q_t S_t^{CB} + q_t B_t^{CB} \leq R_t^k Q_{t-1} S_{t-1}^{CB} + R_t^b q_{t-1} B_{t-1}^{CB} + M_t^B, \quad (44)$$

where T_t^{CB} are seigniorage transfers from (to) the central bank to (from) the Treasury when the former makes profits (losses).²⁰

Government bond and private capital markets clear when the respective (clearing) conditions are satisfied

$$B = B_t^H + B_t^B + B_t^{CB} \quad (45)$$

$$S_t = S_t^H + S_t^B + S_t^{CB}. \quad (46)$$

I gain further insights on the effects of central bank balance sheet policies by inserting the market clearing conditions (45) and (46) into (37) to get an adjusted measure of the aggregate leverage constraint²¹

$$Q_t (S_t - S_t^H) = \phi_t N_t + Q_t S_t^{CB} + \Delta q_t (B_t^{CB} - (B - B_t^H)). \quad (47)$$

Equation (47) implies that when the incentive constraint binds ($\nu_t^{IC} > 0$), [Wallace \(1981\)](#) neutrality theorem breaks, and any increase in security holdings (be it public or private) by the central bank, all else equal, has a positive effect on net private asset demand and asset prices. Elevated asset prices boost bank net worth, suppress excess returns by expanding credit supply, and relax the balance sheet constraint.²²

²⁰I abstract from asymmetries in the transfers of profits to (from) the Treasury ([Hall and Reis, 2016](#)). A similar setup is found in [Cui and Sterk \(2021\)](#) and [Sims et al. \(2022\)](#), among others.

²¹I drop out the ωM_t term because in the calibration I consider it is assumed that ω is zero.

²²Purchases of either asset affect excess returns of both assets since excess returns are related through the arbitrage equation (33).

Another takeaway from the aforementioned equation is that not all assets carry an equal weight in softening intermediaries' leverage constraint. Because it is more likely for a financial intermediary to divert private relative to government assets ($\theta > \theta\Delta$), purchases of the former by the central bank are expected to have a greater impact on total private asset demand and arbitrage opportunities.

Let ϕ_b be the share of total government assets the central bank acquires during *quantitative easing* (*QE*), and ϕ_s the share of central bank purchases of private assets during *credit easing* (*CE*). I can express the quantitative and credit easing rules as

$$B_t^{CB} = \phi_{b,t} B \quad (48)$$

$$S_t^{CB} = \phi_{s,t} S_t, \quad (49)$$

where $\phi_{b,t}$, $\phi_{s,t}$ follow an AR(2) process. It is worth noting that here I have chosen to keep asset purchases exogenous to the model.²³

Treasury The treasury collects lump sum taxes from intertemporal optimizing households ($T_t = (1 - \eta)t_t^o$), uses seigniorage transfers from the central bank (T_t^{CB}), and issues long term government debt in fixed supply, denoted by $B_{t+1} = B_t = B$, to finance government expenditures. Government expenditures comprise government consumption (spending) and the cost of servicing maturing debt. I assume that government spending is constant and accounts for a fraction γ^G of steady state final output

$$G = \gamma^G Y. \quad (50)$$

In addition to taxing optimizers, the government imposes proportional taxes on firms' profits (Div), the proceedings of which are then redistributed to constrained households following an endogenous transfer rule²⁴

$$tr_t = \tau_{div} Div_t. \quad (51)$$

²³Sims and Wu (2021) and Cui and Sterk (2021) postulate endogenous quantitative easing reaction functions, whereby asset purchases respond to different measures of economic activity.

²⁴Those are aggregate transfers to η constrained households.

τ_{div} is the corporate tax rate. Finally, the government budget constraint is expressed as

$$G + R_t^b q_{t-1} B = q_t B + T_t + T_t^{CB}, \quad (52)$$

with T_t being the fiscal instrument that endogenously adjusts to ensure fiscal solvency.

2.6 Long term yields

As is widely discussed in the literature, large scale asset purchases programs (LSAPS) are effective in suppressing term and risk premia through liquidity injections.

In my model, the long term government bond is structured as a consol promising to deliver a constant coupon payment equal to a unit of the final good per period. So, I can infer the yield to maturity using the consol pricing equation

$$q_t = \sum_{s=1}^{\infty} \frac{1}{\left(RL_t^b\right)^s}, \quad (53)$$

where RL_t^b defines the gross yield to maturity on the long term government security.

Similarly, the expected gross payoff of the state-contingent capital claim is given by the marginal product of capital less depreciation. Since private capital is a security with infinite duration (as long term government bonds), the pricing equation becomes

$$Q_t = \sum_{s=1}^{\infty} \frac{Z_{t+s} - \delta}{\left(RL_t^k\right)^s}, \quad (54)$$

and RL_t^k stands for the yield to maturity on the long term private security. Those two equations will be important in determining the macroeconomic transmission mechanisms of unconventional monetary policies explored in the following section.

2.7 Production

There are three types of non financial firms in my model. A perfectly competitive, representative final good producer (retailer) aggregates differentiated intermediates inputs i into a single final good $Y_t = \left(\int_0^1 Y_{i,t}^{\frac{\epsilon-1}{\epsilon}} di\right)^{\frac{\epsilon}{\epsilon-1}}$, where $\epsilon > 0$ is the elasticity of substitution between intermediate varieties. Optimization yields the demand function for each inter-

mediate variety i , given by $Y_{i,t} = \left(\frac{P_{i,t}}{P_t}\right)^{-\epsilon} Y_t$. $P_t = \left(\int_0^1 P_{i,t}^{1-\epsilon} di\right)^{\frac{1}{1-\epsilon}}$ is the aggregate price index. Intermediate goods firms $i \in (0, 1)$ operate under monopolistic competition and produce differentiated goods $Y_{i,t} = A_t K_{i,t}^\alpha H_{i,t}^{1-\alpha}$, using capital and labor. Market power implies that monopolists set prices taking demand as given. Each period, monopolists set the price of their individual variety $P_{i,t}$ to maximize profits subject to a [Rotemberg \(1982\)](#) adjustment cost. Hence, they solve $\max_{\{P_{i,t+k}\}} E_t \sum_{k=0}^{\infty} \Lambda_{t,t+k}^o \left(\frac{P_{i,t+k}}{P_{t+k}} Y_{i,t+k} - \frac{\gamma_p}{2} \left(\frac{P_{i,t+k}}{P_{i,t+k-1}} - \pi \right)^2 Y_{t+k} \right)$ given the demand for individual good i , where $\frac{\gamma_p}{2} (\pi_t - \pi)^2 Y_t$ is the price adjustment cost function and γ_p governs the degree of price adjustment costs. Competitive capital producers buy the final good and any undepreciated capital to invest in the production of new capital goods. Since the supply side follows a standard treatment, I defer the detailed description to [Appendix A.1](#).

Resource constraint. Aggregate resources in the economy are distributed toward household and government consumption, firms' investment, and costs associated with adjustments in prices and wages. Clearing in the goods market implies

$$Y_t = C_t + I_t + G + \frac{\gamma_p}{2} (\pi_t - \pi)^2 Y_t + \frac{\gamma_w}{2} (\pi_t^w - \pi)^2 Y_t. \quad (55)$$

2.8 Calibration

The model is calibrated at a quarterly frequency for the Euro Area. I distinguish the parameters between conventional and model specific. For the conventional parameters I closely follow the New Area Wide Models (NAWM) of the ECB ([Christoffel et al., 2008](#); [Coenen et al., 2018](#)), and standard values in the New Keynesian literature. The five internally calibrated parameters $\{\Delta, \sigma_B, \theta, \kappa_h, \xi\}$ are meant to capture specific moments in E.A. statistics.

I begin with the conventional parameters. The household discount factor is set to $\beta = 0.9950$, allowing for an annualized net real interest rate of 2% in steady state. Capital share and the depreciation rate take common values of $\alpha = 0.36$ and $\delta = 0.025$, respectively. The inverse elasticity of labor supply is fixed to $\varphi = 2.00$, a value widely employed in the literature. The supply of long duration real government bonds $q\bar{B}$ is exogenously set to 70% of annualized steady state real GDP. I align household holdings

of public bonds and private claims with Euro Area long run averages. In particular, I set the fraction of government debt held by households to 75% of the steady state aggregate supply, i.e. $\bar{B}^H / B = 0.75$. The amount of private securities within the household sector is 50% of the corresponding steady state aggregate, that is, $\bar{S}^H / S = 0.50$. Using Euro Area household data from the Eurosystem Household Finance and Consumption Survey, [Slacalek et al. \(2020\)](#) estimates the share of hand to mouth households to account for roughly 20% of the total population. Accordingly, I set $\eta = 0.20$. The portfolio adjustment cost parameter is set to $\kappa = 1.00$ ([Gertler and Karadi, 2013](#)).

Table 1. Parameterization

Parameter	Description	Value	Parameter	Description	Value
Households					
β	Discount factor	0.9950	κ_h	Utility weight of labor	12.896
φ	Inverse Frisch elasticity	2.0000	κ	Portfolio adjustment cost	1.0000
$\bar{B}^H / B$	Share of long term bonds	0.7500	$\bar{S}^H / S$	Share of capital claims	0.5000
η	Fraction of constrained	0.2000			
Financial Intermediaries					
θ	Fraction of divertible assets	0.4473	Δ	Seizure rate of long bonds	0.5000
ω	Seizure rate of reserves	0.0000	ξ	Transfer to new bankers	0.0022
σ_B	Survival rate of bankers	0.9500			
Firms					
α	Capital share	0.3600	δ	Depreciation rate	0.0250
ψ_I	Investment adj.cost	2.5000			
Price and Wage Setting					
ϵ	Goods elasticity of subst.	2.5400	ϵ_w	Labor elasticity of subst.	4.3400
γ_p	Price adjustment cost	108.86	γ_w	Wage adjustment cost	568.48
Treasury					
$q\bar{B}/4Y$	Supply of gov. bonds	0.7000	G/Y	Debt/GDP ratio	0.2000
τ_{div}	Proportional tax on profits	0.2700			
Central Bank					
ϕ_π	Inflation response	1.5260	ϕ_y	Output response	0.0370
ρ_{b_1}	AR(1) coeff. for gov. bonds	1.6800	ρ_{b_2}	AR(2) coeff. for gov. bonds	-0.7300
ρ_{s_1}	AR(1) coeff. for private	1.6800	ρ_{s_2}	AR(2) coeff. for private	-0.7300
	claims on capital			claims on capital	
ρ_m	Interest rate smoothing	0.6500			

The elasticities of substitution for differentiated goods and labor services are parameter-

ized to $\epsilon = 2.54$ and $\epsilon_w = 4.34$ (Coenen et al., 2018). For the inverse elasticity of gross investment to the price of capital, I follow the estimates in Christiano et al. (2005) and set $\psi_I = 2.50$. The price and wage stickiness parameters are chosen to yield the same linear Phillips curves as in the Calvo (1983) framework. This involves $\gamma_p = 108.86$ and $\gamma_w = 568.48$. In the baseline calibration I configure the profit tax rate to $\tau_{div} = 0.27$, which complies with the average corporate tax rate ($\approx 27\%$) in OECD over the period 2000-2018.²⁵ Government consumption is assumed to be 20% of steady state output. Taylor rule inflation and output coefficients take values of $\phi_\pi = 1.526$ and $\phi_y = 0.037$, consistent with the lower bound of Smets and Wouters (2003) estimates. The interest rate smoothing parameter is set to $\rho_m = 0.65$ (Smets and Wouters, 2003).

Now, I turn to the parameters that are specific to my model and calibrated (internally) to match key moments in E.A. data. I calibrate the scaling parameter on the disutility from labor to $\kappa_h = 12.896$ such that steady state labor supply is normalized to $1/3$. The absconding fraction of private capital is calibrated to $\theta = 0.4473$ to target an annualized steady state excess return of capital over the risk free rate ($R^k - R$) of 2.17 percentage points. This is based on the interest rate differential between the long term cost of private borrowing and the EONIA rate from 2003:Q1 to 2014:Q3. I calibrate the proportional seizure rate on long term government bonds to yield an annualized steady state excess return on long term bonds ($R^b - R$) of 1.085 percentage points. This implies that $\Delta = 0.50$. Following data on Euro Area sectoral accounts, Coenen et al. (2018) find the steady state weighted average of financial institutions' and non financial corporations' leverage between 1991:Q1 and 2014:Q4 to be roughly equal to six. Consistent with this target, the fraction of transfers to entering bankers is residually calibrated to $\xi = 0.0022$. Moreover, assuming that bankers stay active for 5 years on average, the survival rate is calibrated to $\sigma_B = 0.95$. To close the banking sector, I set $\omega = 0$ to account for the lack of frictions in the market for reserves. This implies that reserves become fully recoverable by households.²⁶

I model the asset purchase program implemented by the central bank with an AR(2) process. This is the best way to describe the functioning of open market operations the ECB designed amid the recent financial crisis (Andrade et al., 2016; Hohberger et al.,

²⁵Sensitivity analysis indicates that for reasonable values of this parameter ($\tau_{div} \leq 40\%$), the quantitative performance of the model remains unaffected.

²⁶A similar assumption is made by Sims and Wu (2021).

2019), and the subsequent pandemic outbreak. The advantage of the AR(2) specification is that it can generate hump-shaped responses to central bank interventions, capturing anticipation effects of a further stimulus in the future.²⁷ To provide some context, the ECB announced an expansion of its balance sheet commencing in the first quarter of 2015. This expansion involved the injection of 60 billion euros for each of the first four quarters of the program, followed by an increase to 80 billion euros (per quarter) over the next year, with a plan for gradual unwinding thereafter. A similar pattern characterizes the pandemic emergency assistance which peaks a year after its initiation. Consistent with empirical evidence, I calibrate the AR(1) and AR(2) coefficients to $\rho_{b_1} = \rho_{s_1} = 1.68$ and $\rho_{b_2} = \rho_{s_2} = -0.73$, respectively, to target the peak of asset purchases in Q5, and a half-life shock duration of roughly six quarters.

2.9 Solution method

I solve the full non-linear system of equations directly, using Dynare’s perfect foresight solver. In particular, due to the high dimensionality of the model, I exploit the Levenberg-Marquardt mixed complementarity problem (LMMCP) solver. The perfect foresight (deterministic) solver implies that agents form expectations based on full knowledge of the model’s equilibrium conditions and future paths of the exogenous variables, preserving certainty equivalence. Since it abstracts from linearization around a specific point (i.e. the steady state), it allows to take into account important nonlinearities inherent in the model.

In what follows, and unless otherwise stated, the economy is in steady state in $t - 1$. In period t the news of a contemporaneous shock catch agents by surprise. Following the impact period, agents have perfect foresight as all future information is perfectly anticipated and there are no expectation errors.

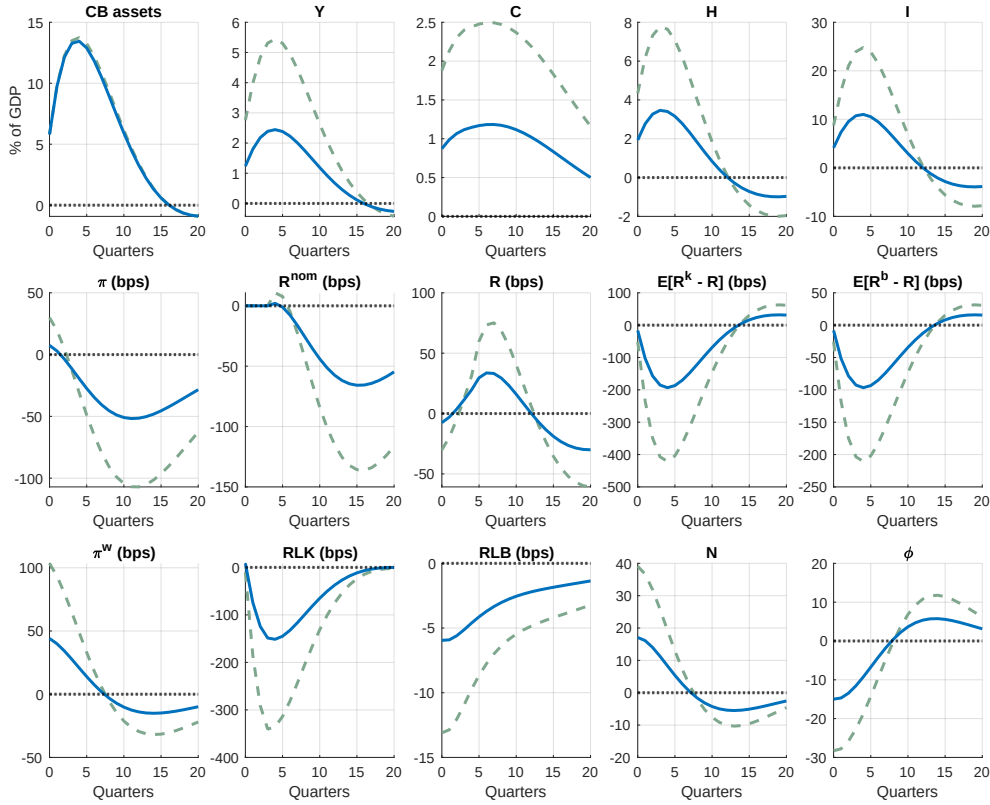
²⁷Under the assumption of an AR(1) process for asset purchases, the peak of the impulse responses happens in the first period, i.e. the period the shock hits. After that, there is a monotonic decay until the economy is back to the initial steady state. This contrasts with the empirical literature, which finds that large scale asset purchases induce delayed peak responses and a gradual return to steady state.

3 SIMULATIONS

In this section, I analyze the buildup of the central bank balance sheet and its quantitative effects on the macroeconomic outlook. I do so by presenting a series of numerical simulations specifically designed to illustrate (i) the differential effects of liquidity interventions in government and private asset markets, along with the channels of the pass-through to (real) aggregates, and (ii) the relative impact on inequality. Then, I examine how a policy reversal reshapes the landscape, focusing on inequality. To isolate the implied strength of the monetary stimulus, I assume no role for fiscal policy.

3.1 Assets purchases and transmission

Figure 4. Asset purchases



NOTES: Blue solid lines: purchases of long term government bonds (quantitative easing); purple dashed lines: purchases of private securities (credit easing). Variables are expressed in percent deviations from steady state. Interest rates, inflation rates, and spreads are expressed in annualized basis point deviations from steady state.

The role of the type of asset purchases Figure 4 illustrates the differential responses of the baseline economy under the two types of security purchases. The solid blue lines refer to central bank purchases of long duration public bonds (*quantitative easing*), and the dashed green lines to the acquisition of long maturity private equity (*credit easing*).

Without loss of generality, I motivate the buildup of the central bank balance sheet focusing on policies adopted during the financial crisis (relative to the pandemic episode). This choice is made to facilitate the comparison between model predictions and empirical evidence. Since the SWIID database provides Gini coefficients only until the quarter just before the onset of Covid-19 and the initiation of the PEPP, it would be impossible to map model estimates against empirical inequality measures due to the lack of relevant post-pandemic data. In terms of real aggregates, the first round of the program (2015:Q1 - 2016:Q1) involved purchases amounting to approximately 7.3% of E.A. real gross domestic product in 2015.²⁸ Breaking this number down implies that the acquisition of government debt instruments accounted for nearly 1.5% of the cumulative purchases, while the remaining 5.8% pertained to purchases of private securities.

Quantitatively, I let the government bond and private capital purchases increase over the course of five quarters, with an initial jump of 5.8% of real output. To replicate the behavior of the ECB in the financial crisis, I postulate a constant policy rate reaction for the first four quarters, after which it endogenously adjusts through an inertial Taylor rule. This resembles the decision also made by the Federal Reserve regarding the funds rate (Chen et al., 2012).

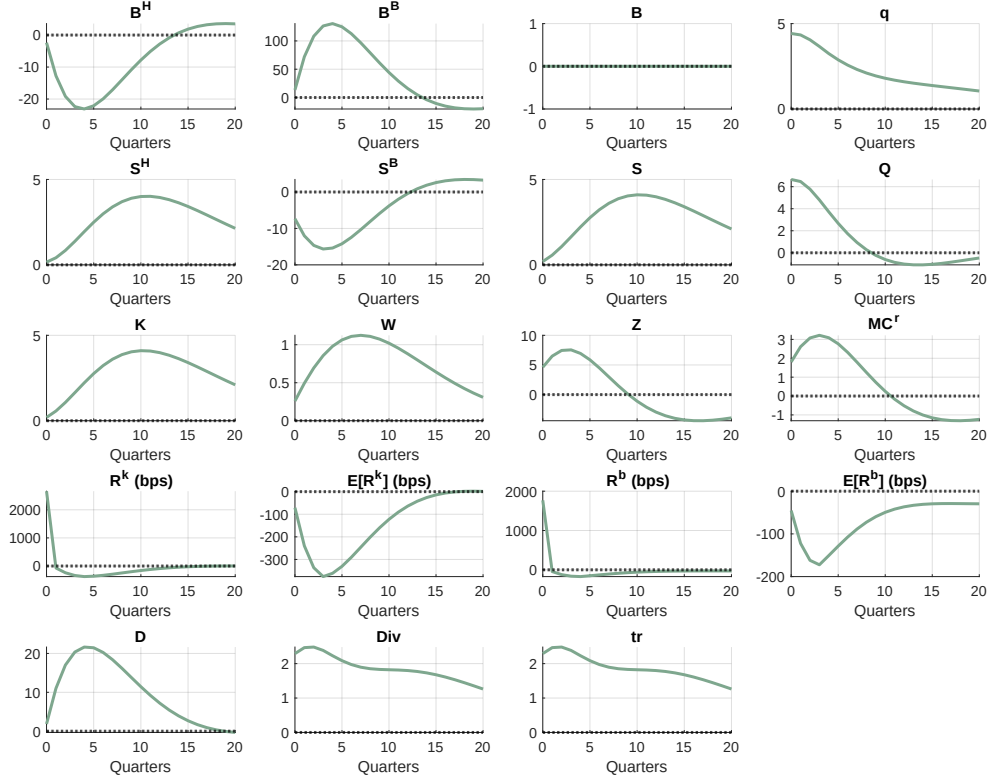
Macroeconomic paths are shown to display a broadly similar qualitative pattern. Quantitatively, purchases of private claims have larger spillover effects on aggregate variables (Gertler and Karadi, 2013). This is because the financial friction is more severe for private assets relative to government bonds. In particular, $\theta > \theta\Delta$ implies that a financial intermediary finds it easier to divert private equity relative to public bonds. Therefore, when the incentive constraint binds, central bank operations in the private equity market increase the total demand for private assets by more than equivalent operations in the market for government debt, *ceteris paribus*.²⁹ This leads to stronger amplification effects.

²⁸Base year is 2010 constant prices.

²⁹One can see this using the expression for the aggregate leverage constraint (47).

Transmission mechanism Having ranked private equity purchases as the critical source of central bank targeted stimulus — relative to purchases of government debt — during the financial crisis, I examine how the former transmits to aggregate variables. Figures 4 and 5 summarize the main impulse responses.

Figure 5. Exogenous credit easing



NOTES: Impulse responses to a 380 basis point shock to purchases of private assets by the central bank. The shock hits in period 0. Variables are expressed in percent deviations from steady state. Interest rates are annualized basis point deviations from steady state.

Credit easing comes with sizeable effects, boosting aggregate output and consumption by 2.7% and 1.9% on impact. When the central bank expands its balance sheet through the acquisition of private claims, it raises the net demand for the asset (i.e. overall demand net the quantity demanded by households, see 47). As a consequence, the price of equity (Q) appreciates and the long term private yield (RL^k) falls (see 54). The fall in the private yield triggers *portfolio rebalancing effects*, as banks substitute private for government securities and households cut down on public debt in exchange for larger equity stakes. Banks' asset substitution is driven by a "*search-for-yield*", while strong wealth

effects steer households equity holdings. Simulations show that the increase in household equity positions does not fully offset the reduction in government bond holdings, with the shortfall being redirected toward higher deposit demand. Overall, total demand for public bonds expands. Given that bonds are in fixed net supply, higher demand inflates bond prices and compresses the yield to maturity on government debt (see 53). Concurrently, elevated asset price valuations raise banks' net worth and suppress excess returns through the arbitrage equation (see 33). This is because higher asset prices depreciate expected returns (see 24, 25) and — for a given real interest rate — excess returns. The decrease in excess returns has two effects; first, it partially offsets the positive impact on banks' valuation from the rise in net worth³⁰; second, it reduces the cost of borrowing, prompting firms to demand more loans. In the aggregate, higher net worth activates a conventional financial accelerator mechanism by making banks less levered, thereby relaxing the incentive constraint. This allows banks to extend lending margins. At the same time, for banks to absorb the excess household deposit demand and satisfy firms' demand for credit, the deposit (risk-free) rate has to decline. Hence, higher supply of credit is accompanied by a drop in the real interest rate, leading to distinct demand and supply effects on real aggregates. On the supply side, greater loan origination is reflected into higher aggregate investment and output. The latter raises demand for intermediate inputs and bids up factor prices (R^k, W). The implied increase in the rental cost of capital together with higher wages raises marginal costs, which pass through to consumers via higher prices, resulting in supply-side inflation. On the demand side, the lower real rate discourages savings and fosters aggregate demand, leading to higher output and demand-side inflation. Overall, demand and supply factors contribute to raising output, consumption, and inflation in response to central bank intermediation.³¹

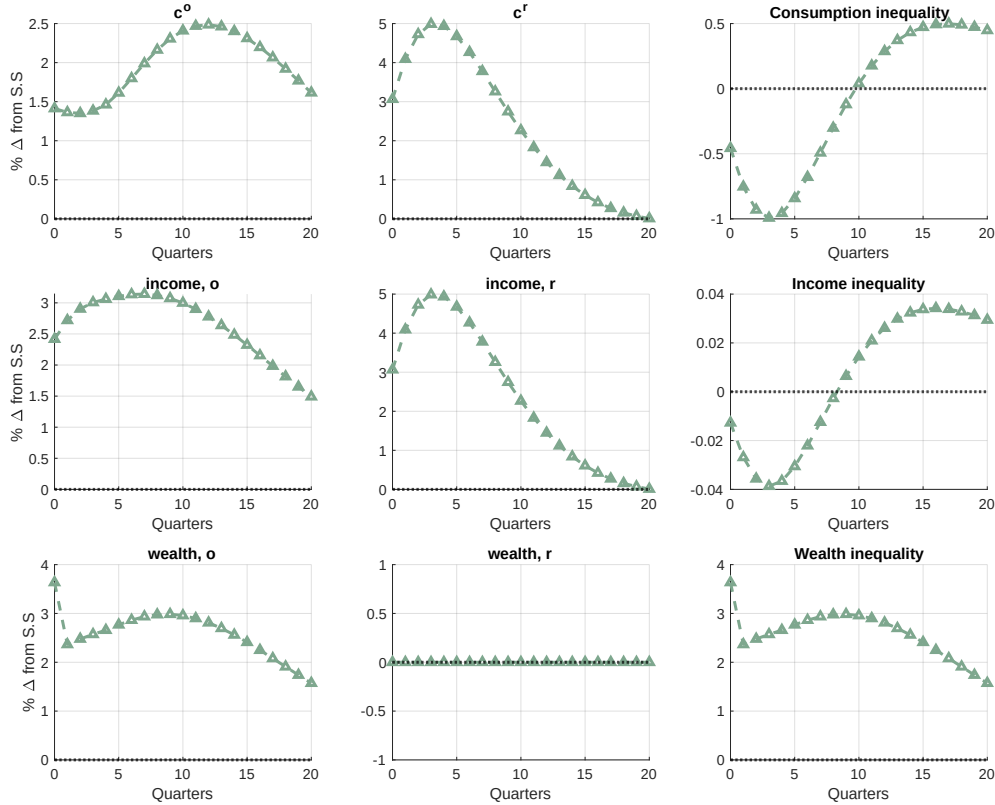
³⁰This can be seen by combining (35), (47), and (27).

³¹Inflation is positive for the first three periods ($t = 0 : 2$ in 4), after which it turns negative. This is due to the forward-looking nature of households in the model. After a strong initial investment (and output) boom following the credit easing, households anticipate real marginal costs to fall. The reason is that higher capital accumulation in the first periods raises the marginal product of labor, inducing firms to reduce labor demand. As such, real marginal costs are expected to slowdown. Iterating forward the standard New-Keynesian Price Phillips curve (A9), a lower path of expected future marginal costs (after $t = 3$) reduces contemporaneous inflation. As the shock fades away, inflation returns to steady state.

3.2 Inequality and redistribution

Figures 6 and 7 jointly account for the consumption, income, and wealth responses of constrained and unconstrained households that ultimately determine the time path of cross-sectional disparities³². Figure 6 shows that credit easing decreases income and consumption inequality, but increases wealth inequality. This corroborates the redistributive role of central bank intermediation, lending support to the view that non-standard expansionary monetary policy reduces income and consumption inequality. To trace out the exact contribution of each group to the observed responses, I perform a careful decomposition of household individual income sources and associate them with standard direct (partial equilibrium) and indirect (general equilibrium) channels of monetary transmission (Kaplan et al., 2018; Auclert, 2019).

Figure 6. Inequality indexes | Credit Easing (CE)

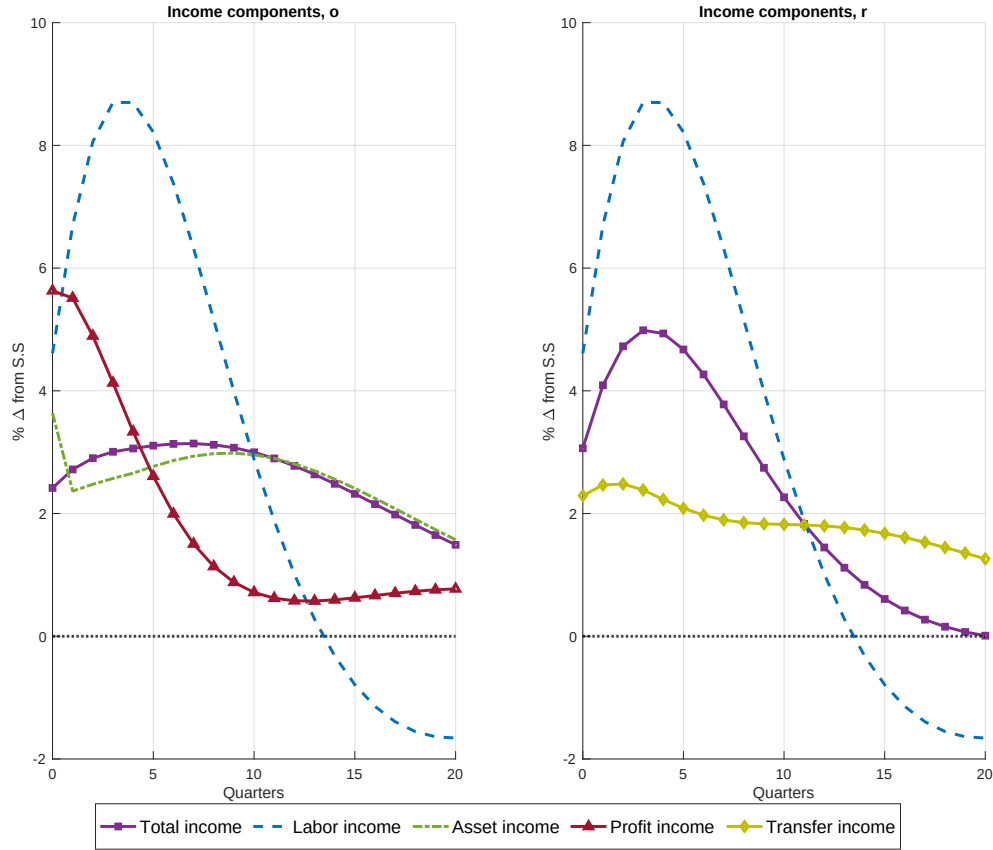


I begin by analyzing the transmission of unconventional monetary policy to the income and consumption of optimizing households. A positive one standard deviation shock to

³²Consumption inequality is defined as the consumption share of $(1 - \eta)$ unconstrained households to total consumption. Similar definitions apply to income and wealth inequality.

central bank purchases of private securities activates at least three channels of transmission. The first is a *portfolio channel*. Credit easing compresses excess returns and boosts asset prices, generating an indirect effect. The rise in asset prices creates a positive wind-fall shock that increases the net worth of banks and firms. Despite the counteracting force of lower discounted expected excess returns, banks — along with firms — benefit from capital gains, leading to higher dividend payouts to optimizers. At the same time, firms experience lower costs of financing capital which steers investment. As investment rises, so does the demand for input factors, pushing up real wages and labor income.

Figure 7. Income sources decomposition



The second is the *savings redistribution channel*, triggered by the decline in the real interest rate. Since unconstrained households are interest rate sensitive, a lower real rate incentivizes them to spend more through intertemporal substitution (direct effect). The increased spending further raises output and labor income. Meanwhile, wage rigidities dampen wage and marginal cost volatility (compared to an environment with flexible wages), causing profits to move with the cycle.³³ In this way optimizers enjoy a jump

³³Bilbiie and Känzig (2023) show that for profits to be procyclical, wages need to be relatively stickier

in profit income. Together, the portfolio and savings redistribution channels shape an *income composition channel*, where higher labor, profit, and asset income, boost the consumption of unconstrained households.

For constrained households, any direct effects are inherently muted, as they are at a kink in their budget set and do not smooth consumption through intertemporal substitution. Higher aggregate demand as a byproduct of the shock raises labor income of constrained households symmetrically to that of the unconstrained. Additionally, transfer income increases via the endogenous transfer rule, which taxes — the now higher — firm dividends and redistributes the proceeds lump sum to hand-to-mouth households. Therefore, general equilibrium forces stimulate the consumption of constrained households through the income composition (indirect) channel.

Overall, credit easing ultimately contributes to narrowing income inequality. This outcome is driven by two key factors. First, the endogenous drop in the deposit rate somewhat dampens the surge in asset income, as households exchange long term government bonds with deposits, when the latter are remunerated at a lower rate than the former.³⁴ Second, when the central bank implements credit easing, government transfers to the central bank rise, creating an imbalance in the government budget constraint. This shortfall is partially settled with higher taxes, which diminish after-tax income of optimizers. As a result, income disparities contract. At the same time, and consistent with the large literature on marginal propensities to consume, constrained households exhibit a stronger consumption reaction compared to the unconstrained, further narrowing consumption inequality. However, as credit easing bolsters asset prices — exclusively benefiting unconstrained households — it simultaneously widens the wealth gap.

Summing up, my findings echo the qualitative insights found in the broad literature while also falling within the spectrum of empirically admissible quantitative estimates. For instance, [Lenza and Slacalek \(2024\)](#) estimates that quantitative easing increases hand-to-mouth consumption by between 1.5% and 3%. In my model, constrained households exhibit a precisely 3% rise in consumption, reinforcing the robustness of the transmission mechanisms identified.

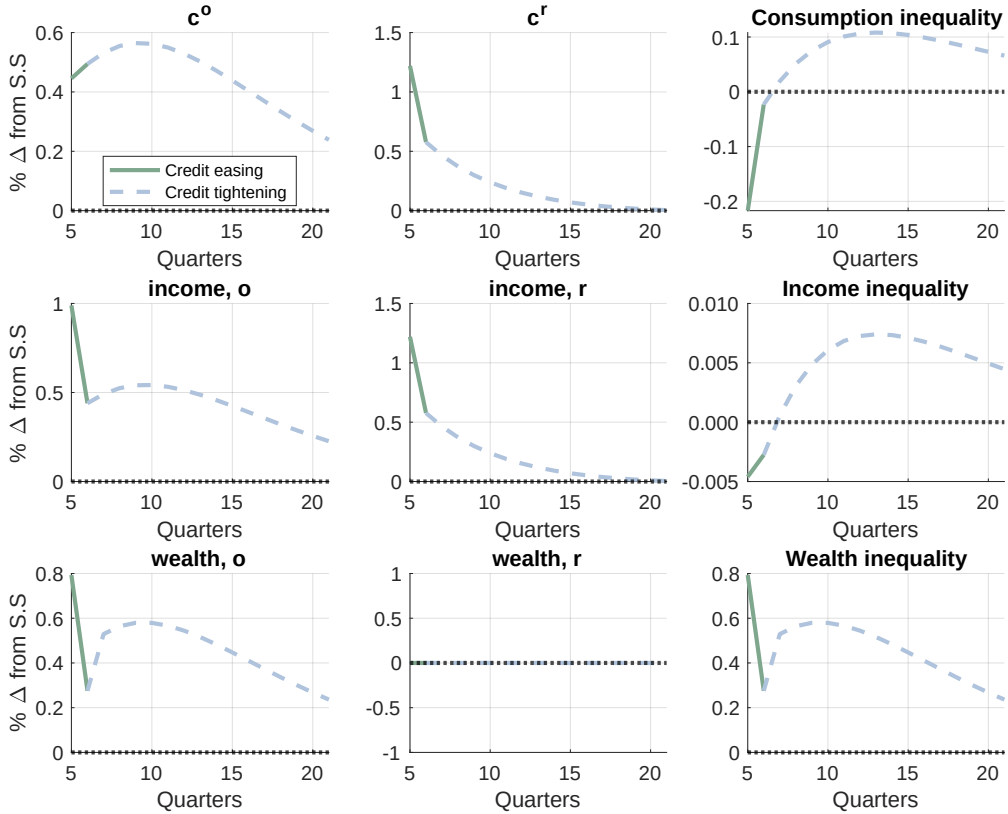
than prices. In my model, wages are (upwardly) more rigid than prices, which is the reason for the cyclicalities of profits.

³⁴Note that asset income contributes the largest share to total income of optimizers, so any variation in the sub-components of asset income have a disproportionally larger effect relative to variations in labor or profit income.

3.3 A tightening cycle

As mentioned in [subsection 3.1](#), scarcity of (post) pandemic data on the evolution of income inequality presents a challenge for properly evaluating the extent to which APP influenced inequality during the pandemic, let alone the strategy reversal that marked the exit. Until now, there has been little evidence on the potential macroeconomic effects of unwinding the central bank balance sheet. Notable exceptions are [Foerster \(2015\)](#) and [Airaudo \(2023\)](#) who study different exit scenarios after a simulated crisis, while [Harrison \(2024\)](#) performs a welfare evaluation of optimal quantitative easing and tightening interventions. Different from the above papers, my focus is on the *distributional dynamics* of a credit tightening, and how this policy initiative shapes the gap between constrained and unconstrained households along the income, consumption, and wealth space.

Figure 8. Inequality indexes | Credit Tightening (CT)



In this respect, figure 8 presents a counterfactual simulation of an aggressive policy reversal characterized by a large — and prompt — unloading of the central bank's balance sheet. My assumption of an aggressive tightening corresponds mainly to the "immediate

sell-off " case of Foerster (2015), and to a less extent, the "aggressive tightening" setting in Airaudo (2023). To construct such a scenario, I let the central bank expand its balance sheet up to quarter five (Q5 is the peak of the CE hump in the data). In quarter six, the central bank sells all its accumulated assets one-off. To make the comparison transparent, the simulation in the graph starts from the last period of credit easing (i.e. Q5), after which the aggressive tightening shock is introduced. As before, during the first four quarters of credit easing the nominal rate is held constant, and then endogenously adjusts via the Taylor rule. A perfect foresight solution is also employed in this experiment.

Income and consumption inequality increase on impact, while the wealth gap contracts.³⁵ This occurs as asset sales drive down asset prices, leading to capital losses for banks. These losses are further exacerbated by declining discounted expected excess returns. At the same time, falling asset prices severely weaken firms' external financing positions, limiting their ability to raise sufficient liquidity for investment. As a result, investment spending and corporate profits decline. The combined capital losses for banks and firms significantly reduce profit income for optimizing households, triggering spillover effects to constrained consumers. Since constrained households rely heavily on transfer income funded by firms' dividends, the decline in corporate profits diminishes these transfers. Additionally, weaker investment reduces total output, lowering demand for production inputs, which in turn suppresses wages and the return on capital. This decline in wages leads to deflationary pressures on goods prices. The fall in real wages activates competing income and substitution effects. In my simulation, the substitution effect dominates, and lower real wages result in a contraction of labor income for both constrained and unconstrained households. Hence, total income for constrained households drops since all its income sources plummet significantly. In a similar vein, asset income of intertemporal optimizers subsides despite the sharp increase in the deposit (risk free) rate due to falling inflation. This is due to the fact that the realized returns on long term corporate and government bonds largely deteriorate, amplifying the negative impact of declining asset prices on their wealth.

Overall, my findings suggest that an aggressive unwinding of the central bank's balance sheet contributes to increasing income and consumption disparities, while the opposite holds true for the wealth gap, which contracts.

³⁵The impact period of credit tightening (CT) is where the dashed lines initiate.

4 CONCLUSION

In this paper I study the impact of the central bank unwinding its balance sheet on various measures of household inequality. To motivate the build up of excess central bank assets and establish a causal theoretical link between monetary policy and inequality — grounded in empirical evidence — I first analyze the effects of expansionary unconventional monetary policy on inequality. I find that a credit easing shock equivalent to 5.8% of real gross domestic product reduces income and consumption inequality, while widening wealth dispersion. I then assess how a policy reversal — similar to the ECB’s post-pandemic strategy shift — reshapes this landscape. My findings highlight a trade-off for monetary policy; while credit tightening helps curb inflationary pressures and aligns with the ECB’s price stability mandate, it widens income and consumption disparities, partly offsetting the equalizing effects of past expansionary policies

Although this paper incorporates several modeling features designed to reflect key aspects of central banking and the broader economy, it has a number of limitations. First, fiscal policy is assumed to be passive, which contrasts with reality — where government interventions played a key role in economic recovery during and after the pandemic, complementing monetary policy. A promising extension would be to endogenize fiscal policy and explore the effects of monetary-fiscal interactions, particularly when fiscal policy is expansionary and monetary policy contractionary. Second, the model abstracts from a key monetary transmission channel — the *earnings heterogeneity channel* ([Ampudia et al., 2018](#); [Komatsu, 2023](#); [Lenza and Slacalek, 2024](#)). Introducing involuntary unemployment through search-and-matching frictions, that can account for differences in job-finding rates and wage dispersion across households, would provide a richer framework to analyze the effects of unconventional contractionary monetary policies on inequality. Finally, an alternative avenue for future research is to compare a gradual tightening cycle, where the ECB unwinds its balance sheet at a steady pace each quarter, against a rapid asset sell-off, and trace out potential differences in the paths of inequality.

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Appendix

A Model

A.1 Production sector

Final good firm A representative final good firm employs intermediate goods $i \in (0, 1)$ to produce a single final good which is then sold to households. The final good firm operates under perfect competition. I assume that each unit of intermediate inputs can produce exactly one unit of final good. It follows that the final output is a CES aggregator of intermediate varieties

$$Y_t = \left[\int_0^1 Y_{i,t}^{\frac{\epsilon-1}{\epsilon}} di \right]^{\frac{\epsilon}{\epsilon-1}},$$

where $Y_{i,t}$ is the intermediate good of producer i .

The final firm chooses the quantity of final output to maximize profits

$$P_t Y_t - \int_0^1 P_{i,t} Y_{i,t} df$$

subject to the demand curve

$$Y_t = \left[\int_0^1 Y_{i,t}^{\frac{\epsilon-1}{\epsilon}} di \right]^{\frac{\epsilon}{\epsilon-1}}.$$

Optimization yields a demand function for each intermediate variety i

$$Y_{i,t} = \left(\frac{P_{i,t}}{P_t} \right)^{-\epsilon} Y_t, \tag{A1}$$

where $P_{i,t}$ is the price of intermediate good i , and P_t the aggregate price index. The zero (aggregate) profit condition for the final good producer

$$Div_t^{FP} = 0 \tag{A2}$$

implies an expression for the aggregate price index

$$P_t = \left[\int_0^1 P_{i,t}^{1-\epsilon} di \right]^{\frac{1}{1-\epsilon}}.$$

Intermediate goods firms Intermediate goods producers span the unit interval $i \in$

$(0, 1)$. Each firm i hires labor and capital to produce its variety according to a constant returns to scale (CRS) production function

$$Y_{i,t} = A_t K_{i,t}^\alpha H_{i,t}^{1-\alpha}, \quad (\text{A3})$$

where A_t is a measure of stochastic aggregate productivity, and α is the capital share. To raise the necessary liquidity to rent capital and hire labor, each intermediate firm i issues $S_{i,t}$ units of capital claims to households and private banks, at the price Q_t . The units of capital claims equal the number of units of acquired capital $K_{i,t+1}$.³⁶ This implies that $Q_t S_{i,t} = Q_t K_{i,t+1}$.

Intermediate firms operate under monopolistic competition and produce differentiated goods. Market power implies that monopolists are price setters. Imposing symmetry, cost minimization yields the demand for labor and capital

$$W_t = MC_t^r (1 - \alpha) \frac{Y_t}{H_t} \quad (\text{A4})$$

$$Z_t = MC_t^r \alpha \frac{Y_t}{K_t}. \quad (\text{A5})$$

Z_t is the rental cost of capital, and MC_t^r the real marginal cost, common across firms and defined as

$$Z_t = R_t^k Q_{t-1} - (1 - \delta) Q_t \quad (\text{A6})$$

$$MC_t^r = \frac{1}{A_t} \left(\frac{W_t}{1 - \alpha} \right)^{1-\alpha} \left(\frac{Z_t}{\alpha} \right)^\alpha. \quad (\text{A7})$$

Monopolists set the price of their individual variety taking demand as given. Price setting is subject to a quadratic adjustment cost (Rotemberg, 1982) when the final price deviates from the target inflation rate π .

The period problem of each monopolist i is to choose the price level $P_{i,t}$ to maximize the present discounted value of real profits

$$\max_{\{P_{i,t+k}\}} E_t \left\{ \sum_{k=0}^{\infty} \Lambda_{t,t+k}^o \left[\frac{P_{i,t+k}}{P_{t+k}} Y_{i,t+k} - \frac{\gamma_p}{2} \left(\frac{P_{i,t+k}}{P_{i,t+k-1}} - \pi \right)^2 Y_{t+k} \right] \right\} \quad (\text{A8})$$

subject to the relative demand for good i (A1). The Lagrange multiplier of the constraint

³⁶Capital needs some time to build, and becomes effective next period.

(shadow price) corresponds to the real cost of producing one extra unit of product i , i.e. $MC_{i,t+K}^r$.

Imposing symmetry as all firms face the same real marginal cost, the profit maximization problem yields a New Keynesian Price Phillips Curve (*NKPPC*)

$$(\pi_t - \pi)\pi_t = \frac{\epsilon}{\gamma_p} \left(MC_t^r - \frac{\epsilon - 1}{\epsilon} \right) + E_t \Lambda_{t,t+1}^o (\pi_{t+1} - \pi)\pi_{t+1} \frac{Y_{t+1}}{Y_t}. \quad (\text{A9})$$

Intermediate producer's i real profits are given by

$$Div_{i,t}^{IP} = \frac{P_{i,t}}{P_t} Y_{i,t} - MC_{i,t}^r Y_{i,t} - \frac{\gamma_p}{2} (\pi_t - \pi)^2 Y_t.$$

Aggregating across intermediate producers implies

$$Div_t^{IP} = \int_0^1 Div_{i,t}^{IP} di. \quad (\text{A10})$$

Capital goods firms. A representative capital good producer owned by optimizers. At the beginning of period t , the capital producer buys the final good (Y_t) and undepreciated capital from intermediate firms $((1 - \delta)K_t)$, and invests to produce new capital goods (K_{t+1}). At the end of the period it sells the newly produced and the refurbished undepreciated capital (K_{t+1}) back to intermediate goods firms at the real price Q_t .

Investment is subject to convex adjustment costs $\Psi(\cdot)$ as in [Christiano et al. \(2005\)](#)

$$\Psi(\cdot) = \Psi\left(\frac{I_t}{I_{t-1}}\right),$$

where $\Psi(1) = 0$, $\Psi'(1) = 0$, $\Psi''(1) > 0$, and I_t stands for gross investment. This is to capture costs associated with changes in investment between time periods. The functional form for the investment adjustment cost is given by

$$\Psi\left(\frac{I_t}{I_{t-1}}\right) = \frac{\psi_I}{2} \left(\frac{I_t}{I_{t-1}} - 1\right)^2 I_t,$$

where ψ_I is the inverse elasticity of gross investment to the price of capital.

The problem of the capital producer is to choose the units of gross investment to maximize the present discounted value of expected future profits

$$\max_{I_t} E_t \sum_{t=0}^{\infty} \Lambda_{t,t+1}^o \left\{ Q_t K_{t+1} - (1 - \delta) Q_t K_t - I_t \right\} \quad (\text{A11})$$

subject to the law of motion of capital

$$K_{t+1} = (1 - \delta) K_t + I_t - \frac{\psi_I}{2} \left(\frac{I_t}{I_{t-1}} - 1 \right)^2 I_t. \quad (\text{A12})$$

The solution to this problem yields a first order condition that determines the price of capital

$$Q_t \left[1 - \frac{\psi_I}{2} \left(\frac{I_t}{I_{t-1}} - 1 \right)^2 - \psi_I \frac{I_t}{I_{t-1}} \left(\frac{I_t}{I_{t-1}} - 1 \right) \right] + E_t \Lambda_{t,t+1}^o \left[Q_{t+1} \psi_I \left(\frac{I_{t+1}}{I_t} \right)^2 \left(\frac{I_{t+1}}{I_t} - 1 \right) \right] = 1. \quad (\text{A13})$$

The Capital producer may earn profits outside of the steady state ([Gertler and Karadi, 2013](#)), given by

$$Div_t^{CP} = Q_t K_{t+1} - (1 - \delta) Q_t K_t - I_t. \quad (\text{A14})$$

It follows that gross aggregate dividends accrued to households (optimizers) from the ownership of non financial firms are the sum of dividends raised by intermediate and capital goods producers

$$Div_t = Div_t^{IP} + Div_t^{CP}. \quad (\text{A15})$$

Accounting for the transfers to hand to mouth households, net dividends accruing to the household sector are expressed as

$$Div_t^H = (1 - \tau_{div}) Div_t. \quad (\text{A16})$$